



BIRLA INSTITUTE OF TECHNOLOGY & SCIENCE, PILANI
Pilani Campus
AUGS/ AGSR Division

FIRST SEMESTER 2026-2027
COURSE HANDOUT (Part II)

Date: 04.08.2026

In addition to the part-I (General Handout for all courses appended to the timetable) this portion gives further specific details regarding the course.

Course No. : **MATH F428**
Course Title : **TIME SERIES ANALYSIS AND FORECASTING**
Instructor-in-charge : **SUMANTA PASARI**
Instructor : **Sumanta Pasari**

1. Course Description:

This course on Time Series Analysis and Forecasting covers introductory concepts of trend, seasonality, and cyclicity in time series data. It reviews extrapolative and decomposition models and smoothing techniques. Discussions include autoregressive error structure – auto-covariance and auto-correlation, ACF, and PACF. Linear stationary model analysis, such as AR, MA, ARMA is included along with error diagnostics. It includes Box-Jenkins time series analysis and linear non-stationary models, such as ARIMA, SARIMA, and ARFIMA. Finally, we study some models with stochastic variance – ARCH and GARCH. The course includes discussions on multivariate time series models and associated techniques. It necessarily involves the use of statistical software and programming languages for hands-on practice.

2. Scope and Objective of the Course:

This 4th level course introduces time series analysis and associated forecasting techniques, in which students will be equipped with various forecasting techniques and knowledge on modern statistical methods for analyzing time series data. The course reviews extrapolative and decomposition models, introduction to Box–Jenkins time series analysis, and ARIMA models. The remainder of the course considers various practical aspects of the principles behind modern forecasting techniques. Students will study and develop the ability to select appropriate model to carry out forecasting. A one-hour lab will be conducted every week. Students will learn how to explore and analyze different types of time series data using R programming.

3. Text Book:

T1: Box, G. E., Jenkins, G. M., Reinsel, G. C., & Ljung, G. M. (2015). Time Series Analysis: Forecasting and Control. John Wiley & Sons..

4. Reference Books:

R1: Diebold, F. X. (1998). Elements of Forecasting. South-Western College Pub..

R2: Krispin, R. (2019). Hands-On Time Series Analysis with R: Perform Time Series Analysis and Forecasting using R. Packt Publishing Ltd..

R3: Shumway, R. H., Stoffer, D. S., & Stoffer, D. S. (2000). Time Series Analysis and Its Applications (Vol. 3). New York: springer.



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5. Course Plan:

A. Lecture (Theory)

Learning Objectives	Topics to be Covered	Lecture Nos.	Ref. To Textbook
Introduction and Overview	Forecasting in Action, Forecasting Methods, Objectives of Time Series Analysis, Some Simple Time Series Models, Stationary Models and the Autocorrelation Function, Estimation and Elimination of Trend and Seasonal Components, Testing the Estimated Noise Sequence	1-4	Chapter 1 of T1+Chapter1 of R1 class notes
A Brief Review of Probability, Statistics, and Regression for Forecasting	Random Variables, Distributions, and Moments, Multivariate Random Variables, Interpreting distributions and densities, Covariance and correlation, Conditional mean and variance, Scatterplots and regression lines, Simple linear regression, Nonlinear least squares	5-7	Chapter 2 of R1+ class notes
Statistical Graphics for Forecasting	Simple Graphical Techniques, Elements of Graphical Style, Application: Graphing Four Components of Real GDP	8-9	Chapter 4 of R1+ class notes
Autocorrelation Function of Stationary Processes	Autocorrelation Properties of Stationary Models, Time Series and Stochastic Processes, Stationary Stochastic Processes, Autocovariance and Autocorrelation Functions, Estimation of Autocovariance and Autocorrelation Functions	10-11	Chapter 2 of T1+ + class notes
Linear Stationary Models Analysis	General Linear Process, Two Equivalent Forms for the Linear Process, Autocovariance Generating Function of a Linear Process, Stationarity and Invertibility Conditions for a Linear Process, Autoregressive and Moving Average Processes, Autoregressive Processes, Stationarity Conditions for Autoregressive Processes, The First-Order Autoregressive Process, Second-Order Autoregressive Process, Partial Autocorrelation Function, Moving Average Processes.	12-17	Chapter 3 of T1+ class notes
Linear Nonstationary Models	Autoregressive Integrated Moving Average Processes, Nonstationary First-Order Autoregressive Process, General Model for a Nonstationary Process Exhibiting Homogeneity, General Form of the ARIMA Model, Three Explicit Forms for the ARIMA Model, Difference Equation Form of the Model.	18-23	Chapter 4 of T1+ class notes
Forecasting	Minimum Mean Square Error Forecasts and Their Properties, Calculating Forecasts and Probability Limits, Forecast Function and Forecast Weights, Eventual Forecast Function Determined by the Autoregressive Operator, Role of the Moving Average Operator in Fixing the Initial Values, Examples of Forecast Functions and Their Updating, Forecasting an IMA(0, 1, 1) Process, Forecasting an IMA(0, 2, 2) Process, Forecasting a General IMA(0, d, q) Process, Forecasting Autoregressive Processes, Forecasting a (1, 0, 1) Process, Forecasting a (1, 1, 1) Process.	24-29	Chapter 5 of T1+ class notes



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Evaluating and Combining Forecasts	Evaluating a Single Forecast, Evaluating Two or More Forecasts: Comparing Forecast Accuracy, Forecast Encompassing and Forecast Combination, Application	30-34	Chapter 12 of R1+ class notes
Tests for Unit Roots in ARIMA Models	Tests for Unit Roots in AR Models, Extensions of Unit Root Testing to Mixed ARIMA Models	35-38	Chapter 10 of T1+ class notes
Conditional Heteroscedastic Models	Introduction to ARCH and GARCH models	39-40	Chapter 10 of T1+ class notes

B. Practical – Lab

Experiment Plan

Week	Module	Experiment Title
1	Time series graphics	Time plots, Time series patterns, Seasonal plots, Seasonal subseries plots, Scatterplots, Lag plots, Autocorrelation, White noise, Exercises
2	Time series decomposition	Transformations and adjustments, Time series components, Moving averages, Classical decomposition, Methods used by official statistics agencies
3	Time series features	Some simple statistics, ACF features, Exploring Australian tourism data, Exercises
4-5	The forecaster's toolbox	A tidy forecasting workflow, Some simple forecasting methods, Fitted values and residuals, Residual diagnostics, Distributional forecasts and prediction intervals, Forecasting using transformations, Forecasting with decomposition, Evaluating point forecast accuracy, Evaluating distributional forecast accuracy, Time series cross-validation, Exercises.
6	Time series regression models	The linear model, Least squares estimation, Evaluating the regression model, Some useful predictors, Selecting predictors, Forecasting with regression, Nonlinear regression, Correlation, causation and forecasting, Matrix formulation, Exercises.
7	Exponential smoothing	Simple exponential smoothing, Methods with trend, Methods with seasonality, A taxonomy of exponential smoothing methods, Estimation and model selection, Forecasting with ETS models, Exercises.
8-9	Forecasting with ARIMA Models	The stationary state of time series data, The random walk process, The AR and MA processes, The ARMA and ARIMA models, The seasonal ARIMA model; Linear regression with the ARIMA errors model.

6. Evaluation Scheme:

Component	Weightage	Date	Nature
Mid-Sem Examination	25%	October 10, 2026	Closed Book
Laboratory Components	15%	Continuous assessment	Open Book
Group Assignments (two)	10%	Will be announced later	Open Book
Quizzes (best three out of four)	15%	Announced	Closed Book
Comprehensive Examination	35%	December 16, 2026	Open/Closed Book

7. Chamber Consultation Hours: To be announced in class



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8. Notices: All notices in relation to the above course will be displayed on NALANDA.

9. Make up policy: Make up for midsem and comprehensive examination will be granted only in genuine cases. **There is no make-up for other components.** Students should follow proper guideline for applying make ups.

Instructor in Charge

MATH F428